databricks

Databricks Unity Catalog

Unified governance for data and AI

Date

Name



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Data and AI governance drives business value

"Organizations are finally realizing the value of data as an asset that needs to be protected, managed and maintained to increase asset value"

—

IDC

"Organizations seeing the highest returns from AI have a framework for AI governance to cover every step of the model development process"

—

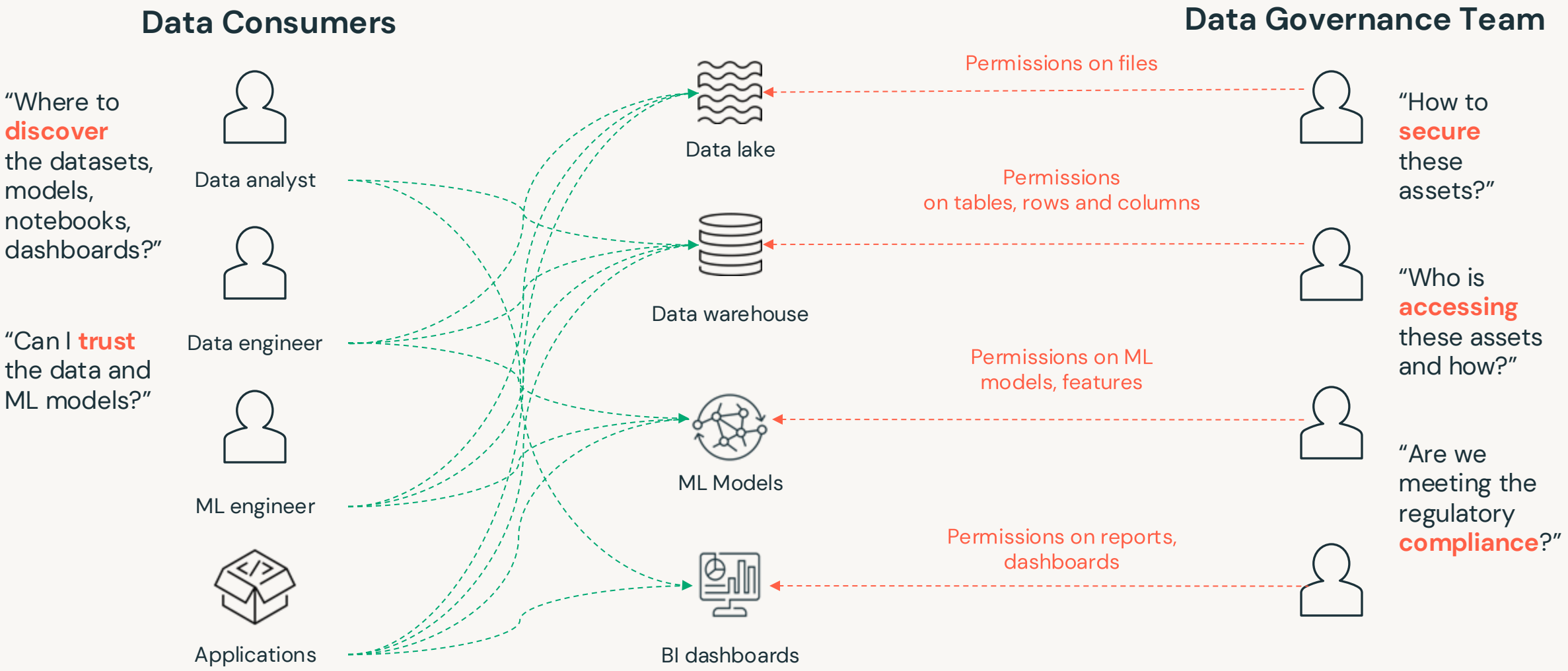
The State of AI in 2022, McKinsey & Co

"AI is now an enterprise essential, and as such, AI governance will join cybersecurity and compliance as a board-level topic"

—

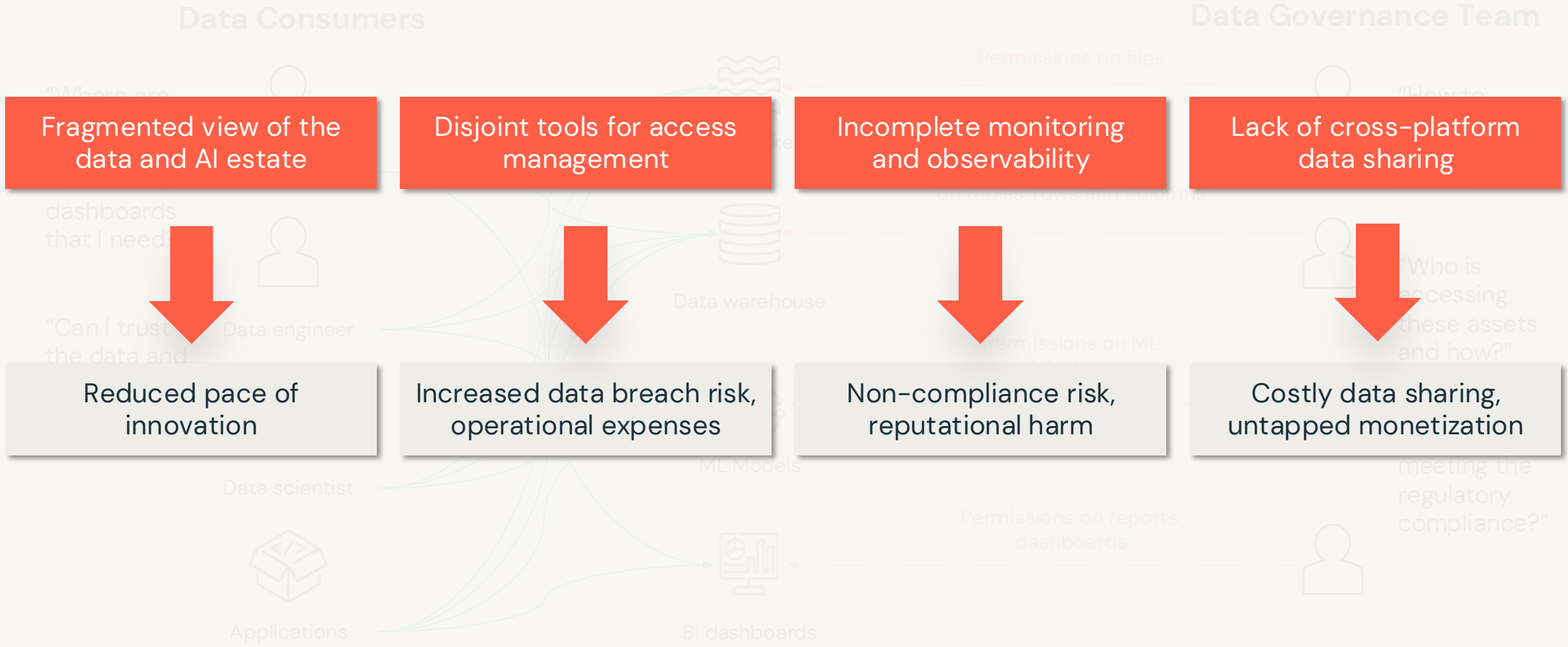
Forrester, 2023 AI Predictions report

Today, data and AI governance is complex



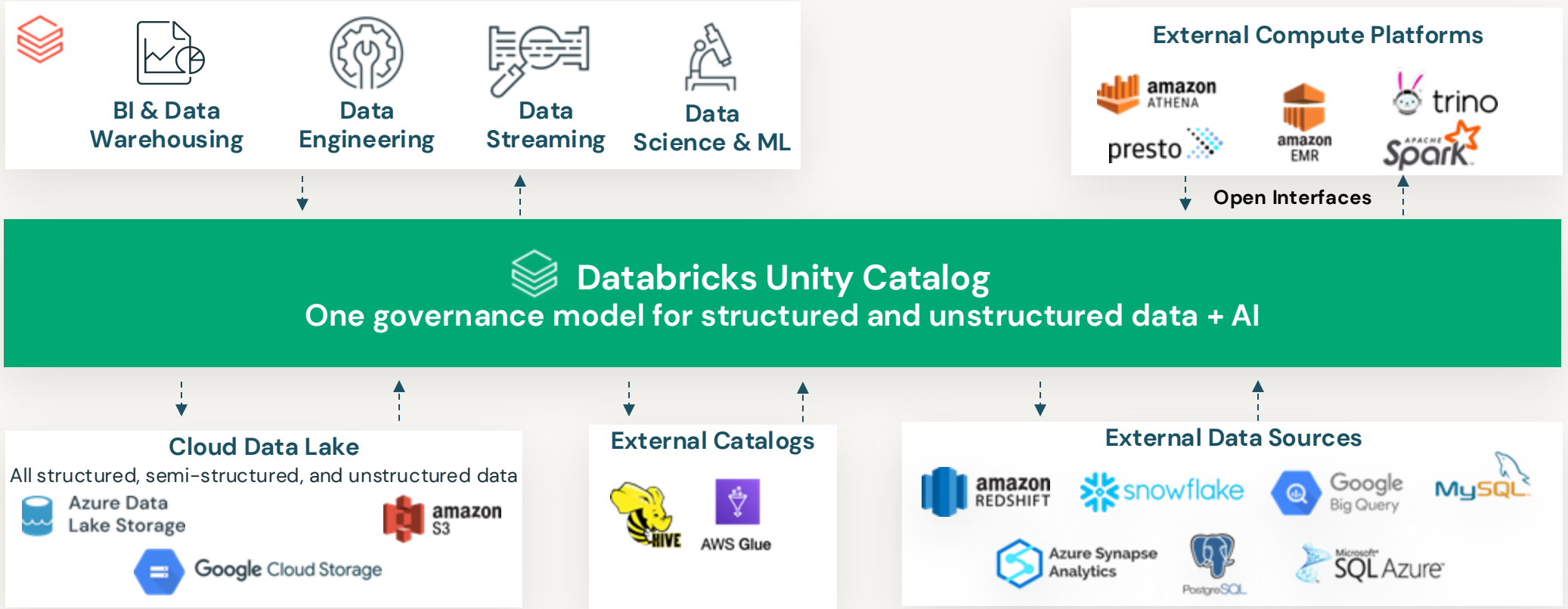
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Today, data and AI governance is complex



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Databricks Data Intelligence Platform unifies data and AI governance



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Databricks Unity Catalog

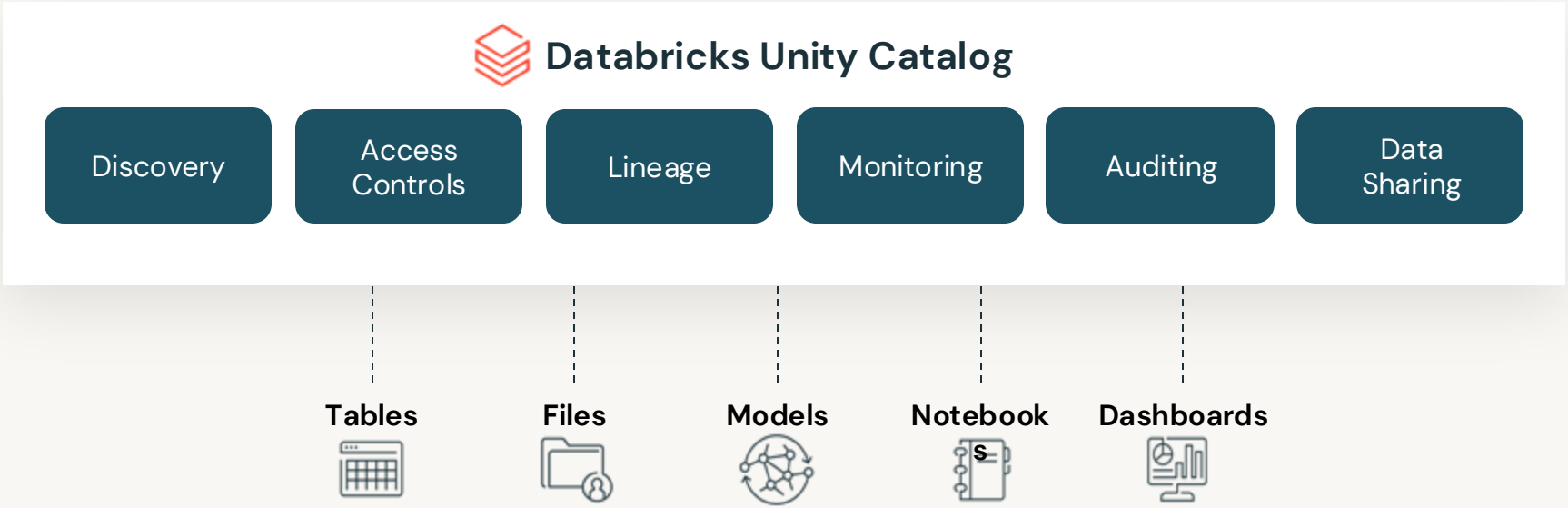
Unified governance for data and AI

Unified visibility into data and AI

Single permission model for data and AI

AI-powered monitoring and observability

Open data sharing



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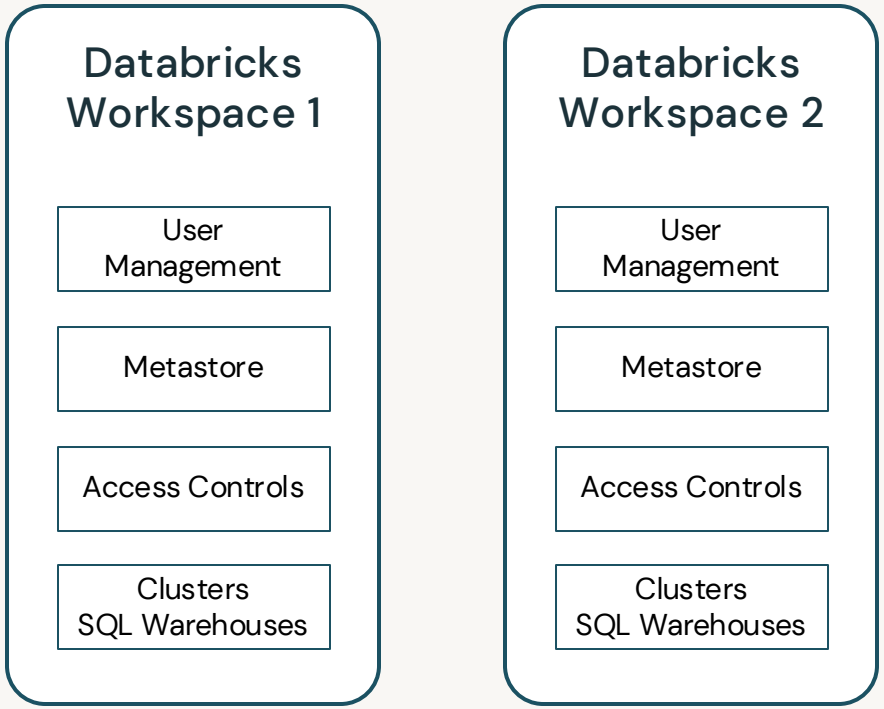
Unity Catalog Unlocks Full Potential of your Lakehouse	w/ UC	w/ Hive Metastore
Governance across workspaces (Access controls, metastore, Identity management)	✅	❌ (Governance at workspace level)
Automated table and column level lineage in real-time	✅	❌
Row Filters and Column Masking (use standard SQL functions for fine-grained access controls)	✅	❌
Volumes (Govern non tabular data/arbitrary files - image, video, PDF etc.)	✅	❌
Lakehouse Federation (Discover, govern and query external databases, data warehouses, Hive Metastore, AWS Glue)	✅	❌
Lakehouse Monitoring (Monitor quality of data and ML models with auto generated alerts and dashboards)	✅	❌
Delta Sharing (Share files, tables, ML models, notebooks across clouds, regions and data platforms)	✅	❌
Databricks Marketplace	✅	❌
Databricks Cleanroom (Collaboration on data in a privacy-preserving manner)	✅	❌
Databricks Data Room (aka Genie)	✅	❌
Databricks Assistant / LakehouseIQ (Knowledge engine on the lakehouse to get insights in natural language)	✅	❌ (Degraded experience)
Governance for AI (manage and govern ML Models, Feature store)	✅	❌
System Tables (end to end observability for Billing, Auditing, Lineage, Marketplace analytics)	✅	❌
Lakehouse Apps	✅	❌
Databricks Connect V2	✅	❌
Materialized Views	✅	❌
AI Powered Predictive optimization (optimizes your query plans and data layout for peak performance, intelligently balancing read vs. write performance)	✅	❌
Streaming Tables	✅	❌
HMS Interface (Query data registered in UC using other data platforms such as Amazon Athena, Presto, Trino,EMR)	✅	✅
Vector Search for indexing	✅	❌
Serverless workflows	✅	❌

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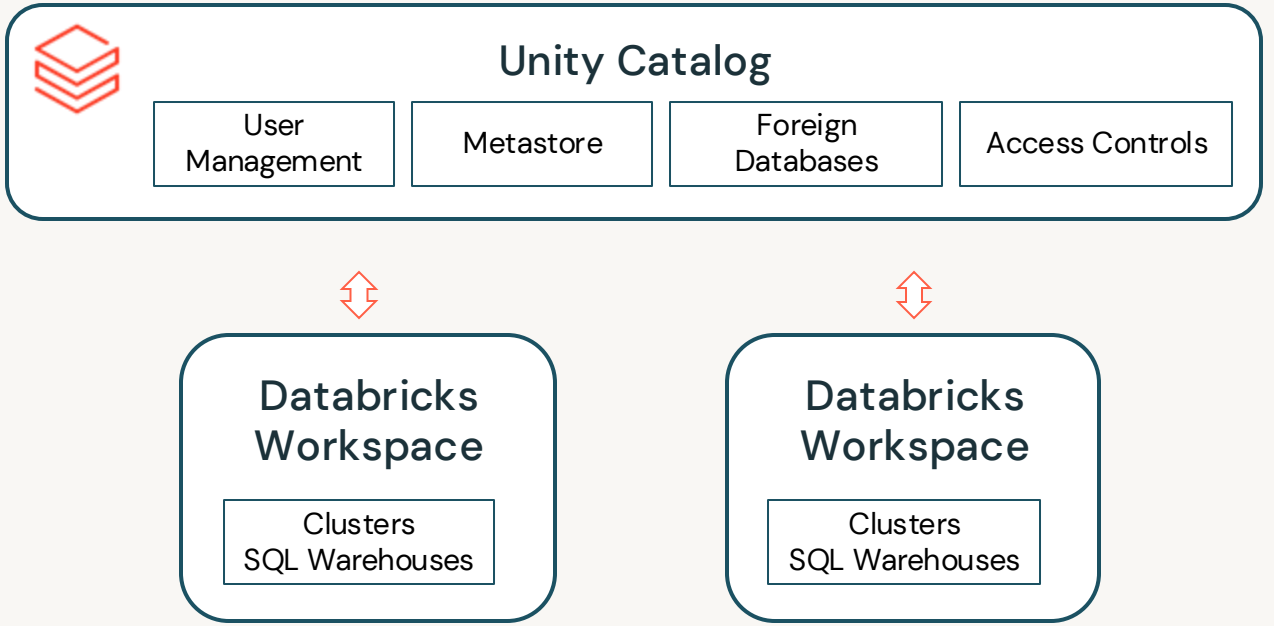
All your metadata, in one place

One metadata layer across file and database sources **superpowers** governance

Without Unity Catalog



With Unity Catalog



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Fundamental Concepts

Working with file based data sources

- Credentials
  - Cloud provider credential to connect to storage
- External Locations
  - Storage location used for external tables or arbitrary files
- Managed Data Sources
  - External Location that is used exclusively for tabular data
- Volumes
  - Arbitrary file container inside an external location

Working with databases

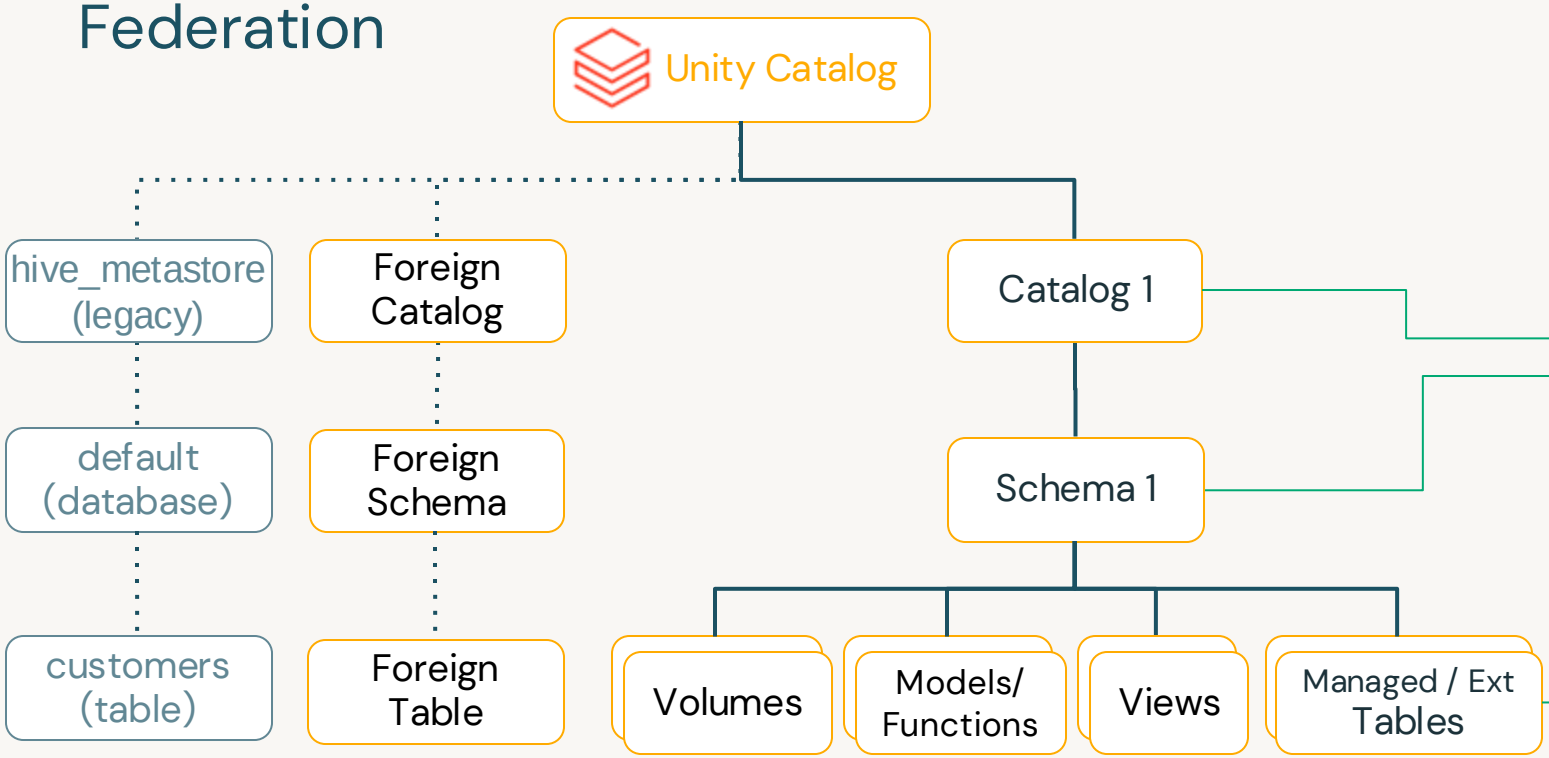
- Connections
  - Credential and connection information to connect to an external database
- Foreign Catalogs
  - A catalog that represents an external database in UC and can be queried alongside managed data sources and file sources

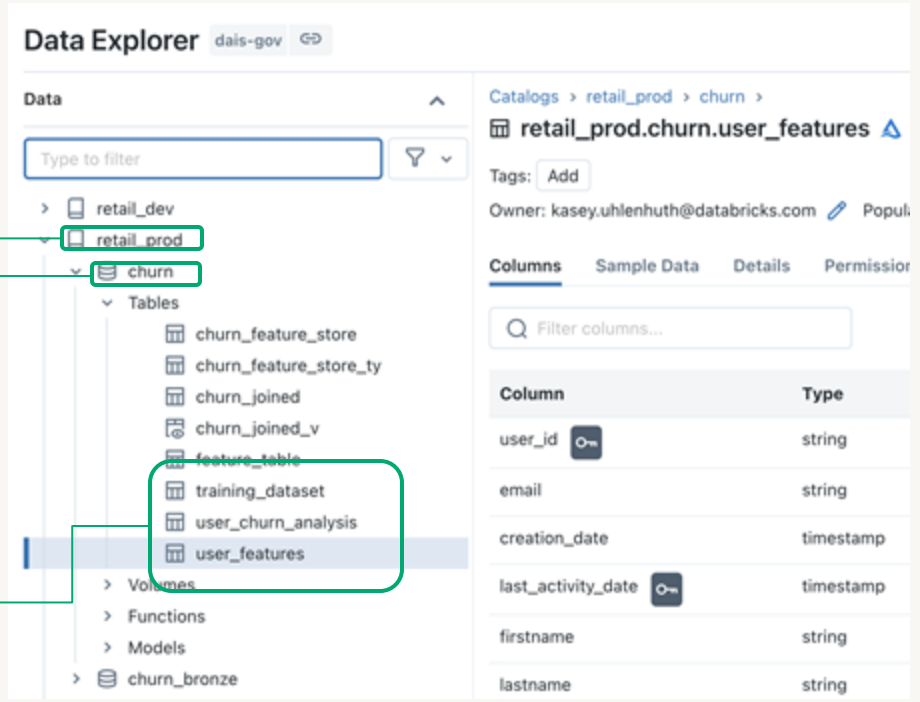
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Governed namespace across file and database sources

Access legacy metastore and foreign databases powered by Lakehouse Federation






```
SELECT * FROM main.paul.red_wine;  -- <catalog>.<database>.<table>

SELECT * FROM hive_metastore.default.customers;

SELECT * FROM snowflake_warehouse.some_schema.some_table;
```

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Centralized Access Controls

Centrally grant and manage access permissions across workloads

Using ANSI SQL DCL

```
GRANT <privilege> ON <securable_type>
<securable_name> TO `<principal>`

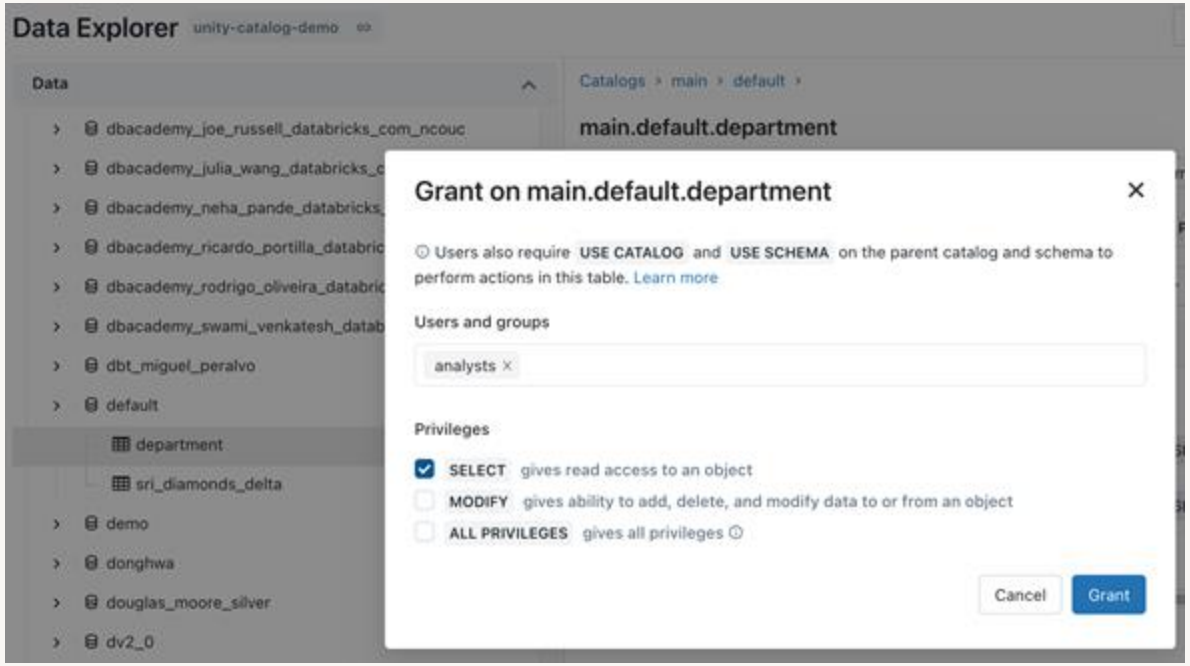
GRANT SELECT ON iot.events TO engineers
```

Choose permission level

'Table'= collection of files in S3/ADLS

Sync groups from your identity provider

Using UI



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Row Level Security and Column Level Masking

Provide differential fine grained access to datasets

Only show specific rows

```
CREATE FUNCTION <name> ( <parameter_name >
<parameter_type> .. )
RETURN {filter clause whose output must be a boolean}
```

```
CREATE FUNCTION us_filter(region STRING)
RETURN IF(IS_MEMBER('admin'), true, region="US");

ALTER TABLE sales SET ROW FILTER us_filter ON region;
```

Test for group membership

Assign reusable filter to table

Specify filter predicates

Mask or redact sensitive columns

```
CREATE FUNCTION <name> (<parameter_name>,
<parameter_type>, [, <column>...])
RETURN {expression with the same type as the first
parameter}
```

```
CREATE FUNCTION ssn_mask(ssn STRING)
RETURN IF(IS_MEMBER('admin'), ssn, "****");

ALTER TABLE users ALTER COLUMN table_ssn SET MASK
ssn_mask;
```

Test for group membership

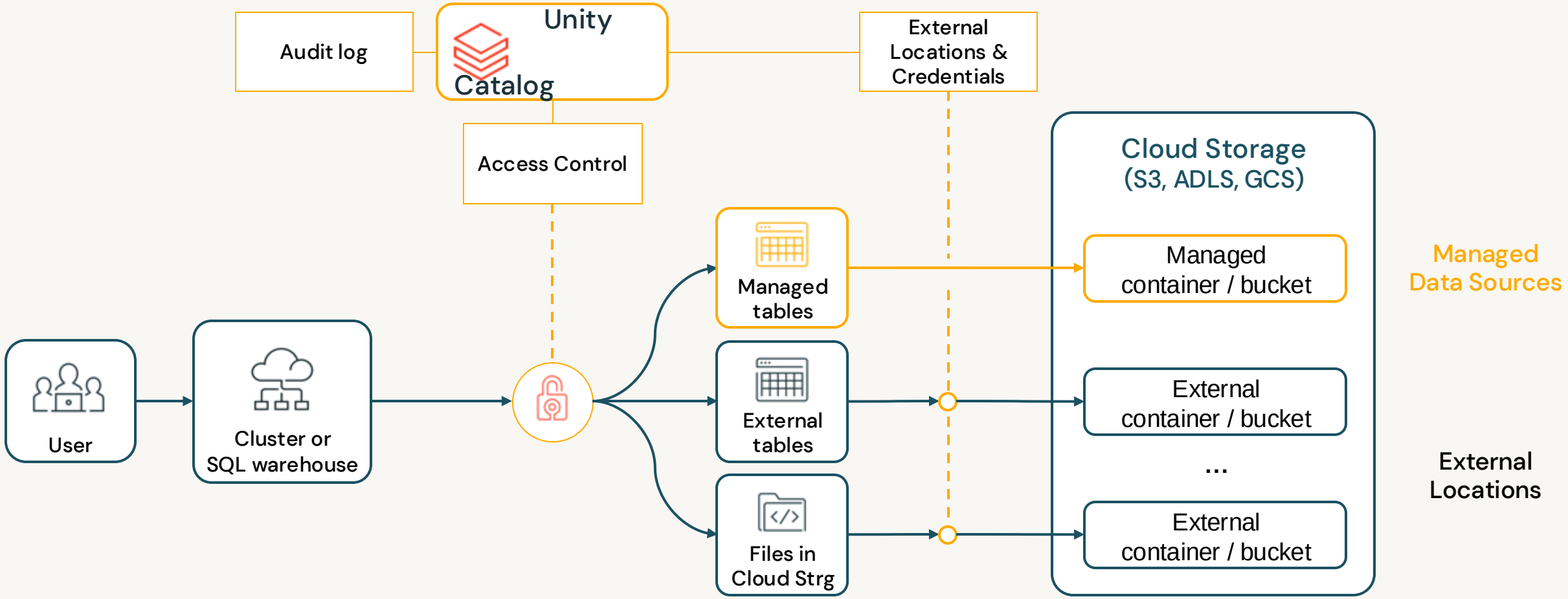
Assign reusable mask to column

Specify mask or function to mask

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Managed Data Sources & External Locations

Simplify data access management across clouds



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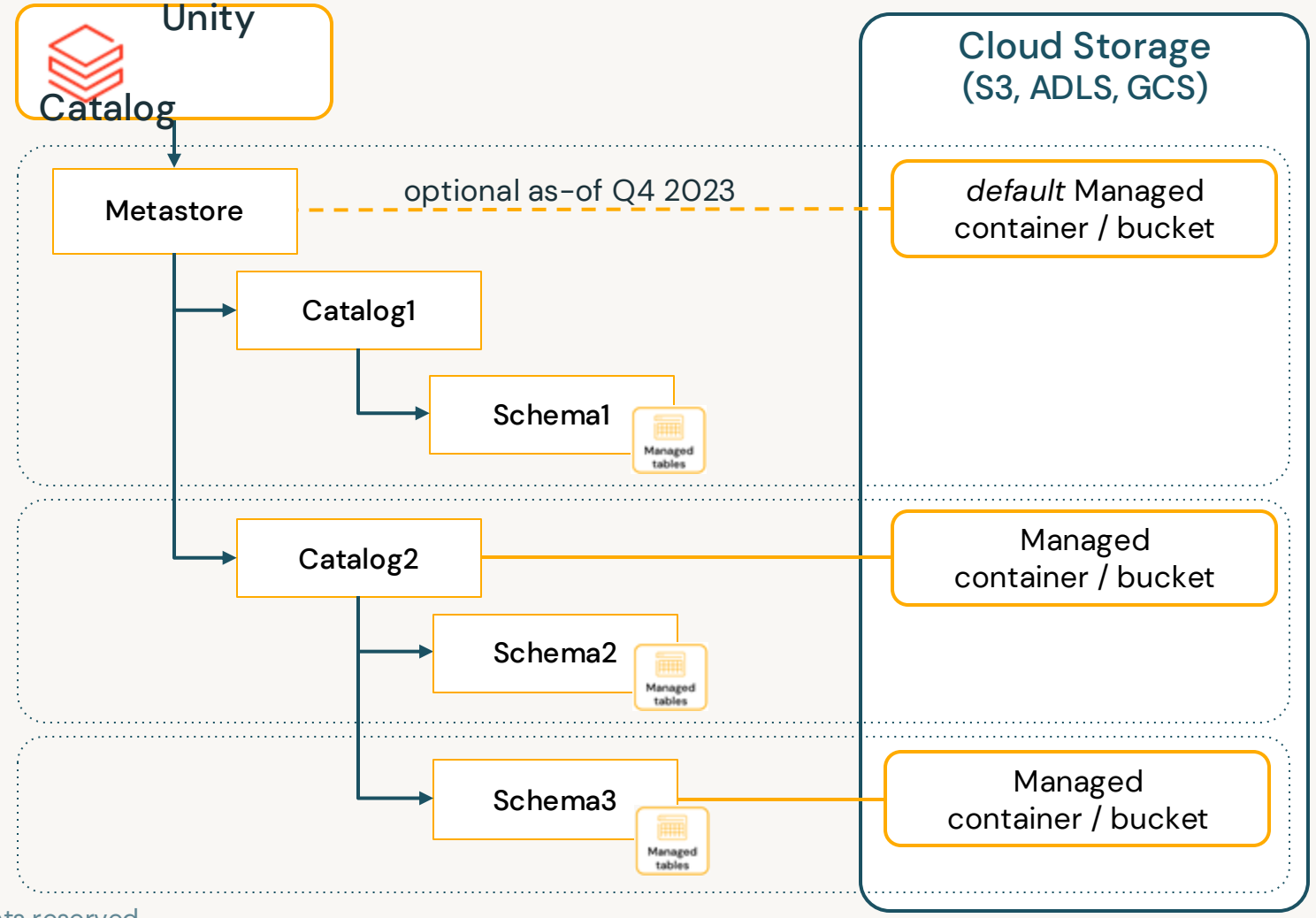
Default access to storage by catalog or schema

Use managed data sources for data isolation or cost allocation

1) default storage associated with the **metastore**

2) define at the **catalog**

3) define at the **schema**



strong recommendation: do *not* store catalogs/schemas in the metastore-level default location

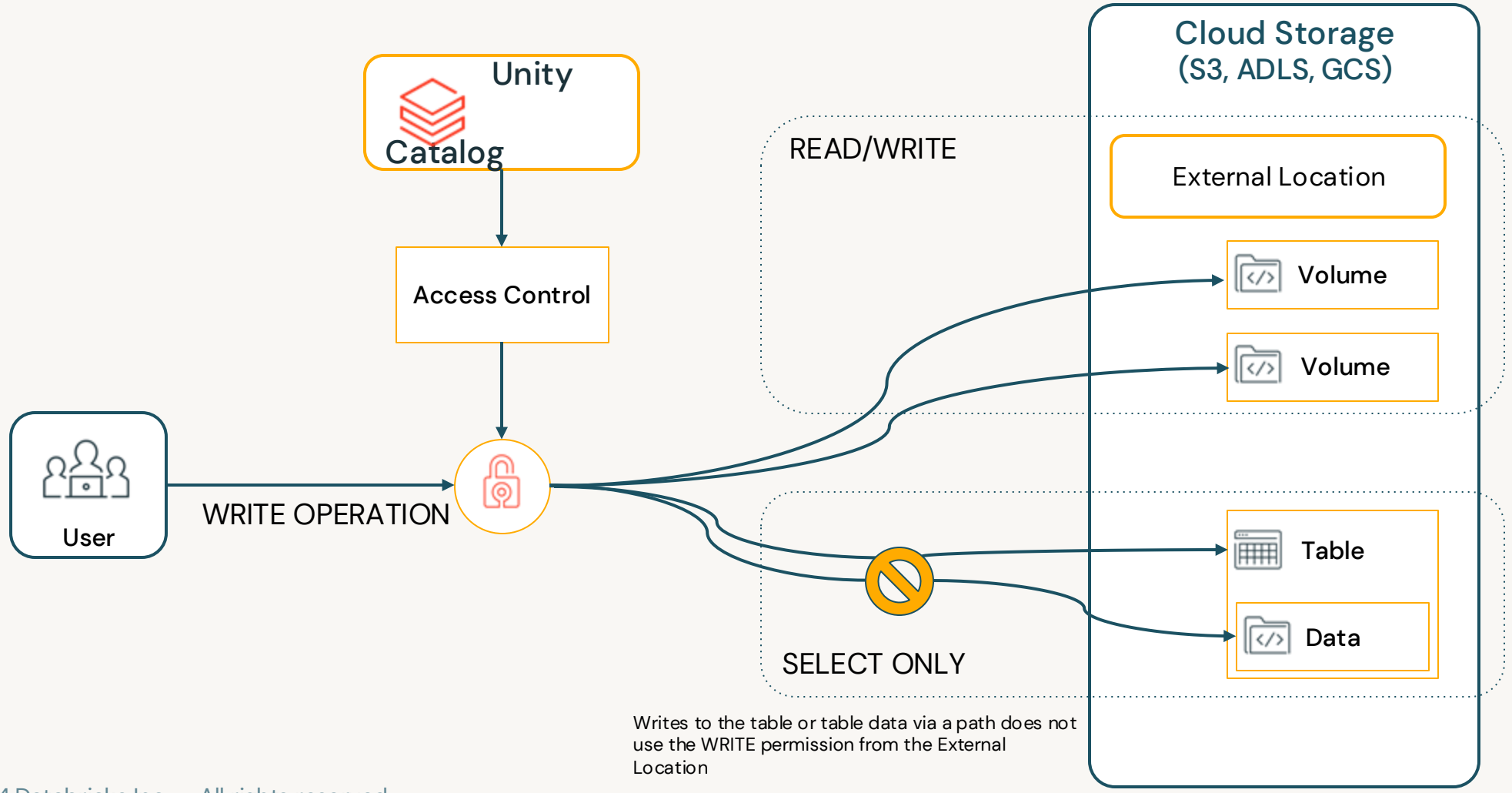
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rather, specify a separate managed storage location for each catalog (which becomes the default location for schemas/tables w/in that catalog)

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Govern filesystems and objects distinctly

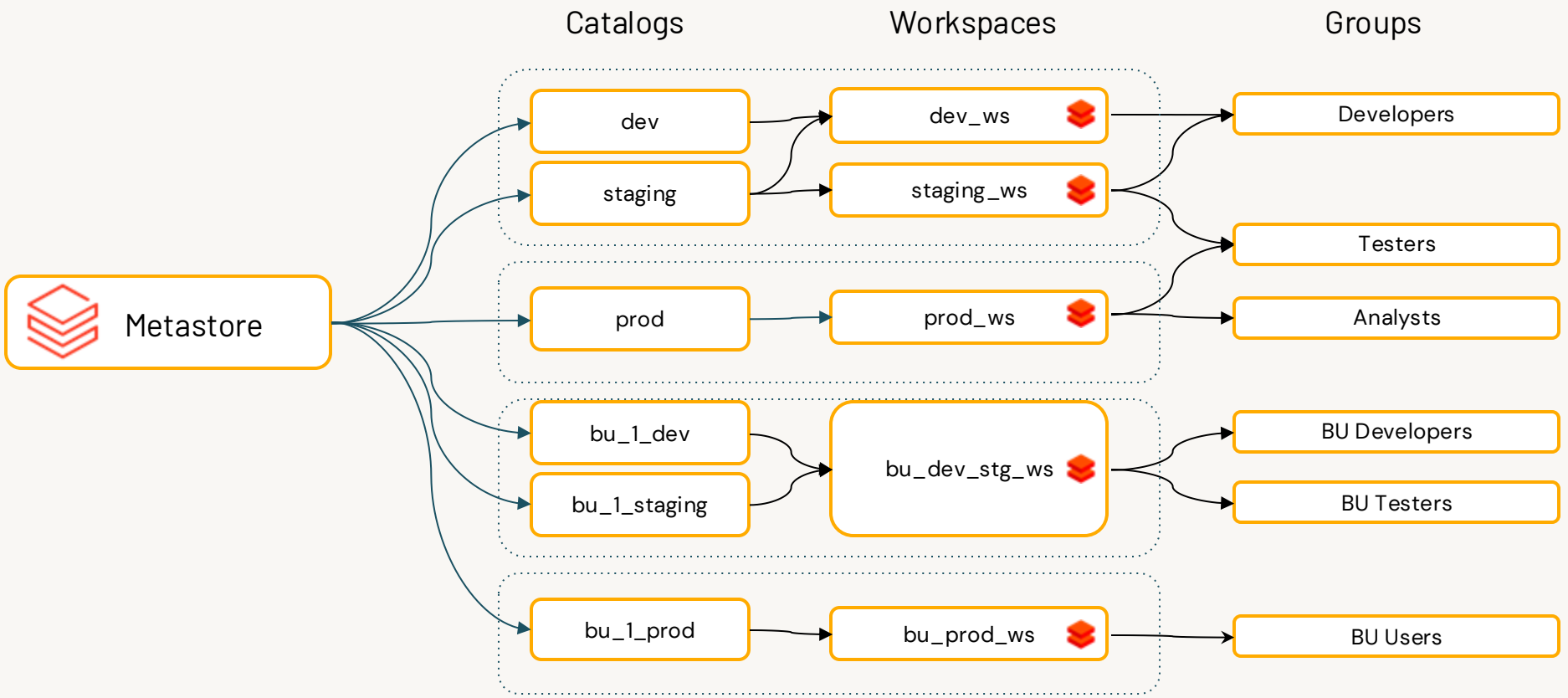
Govern external tables and filesystem access separately



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Access data from specified environments only

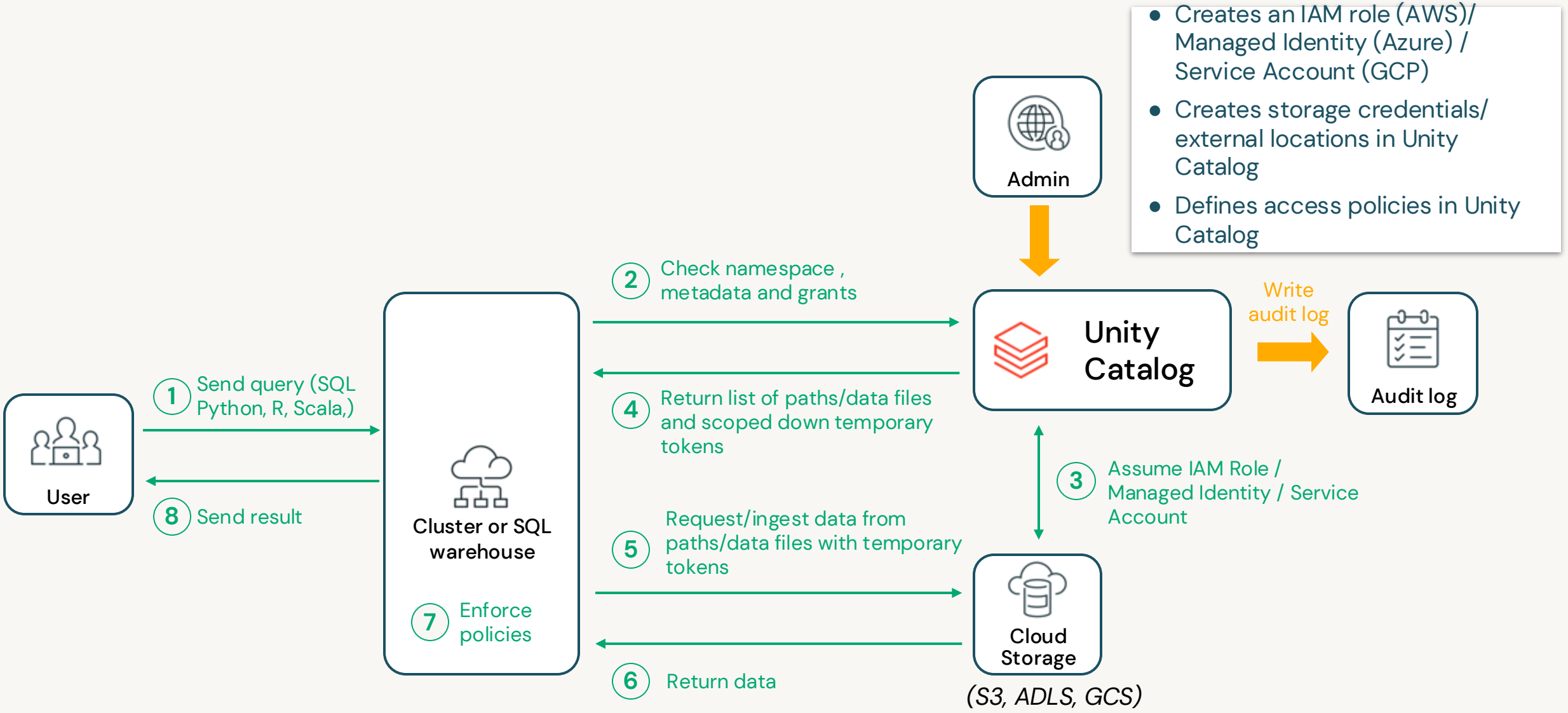
Restrict data access by environment or purpose



Access to data and availability of data can be isolated across workspaces and groups

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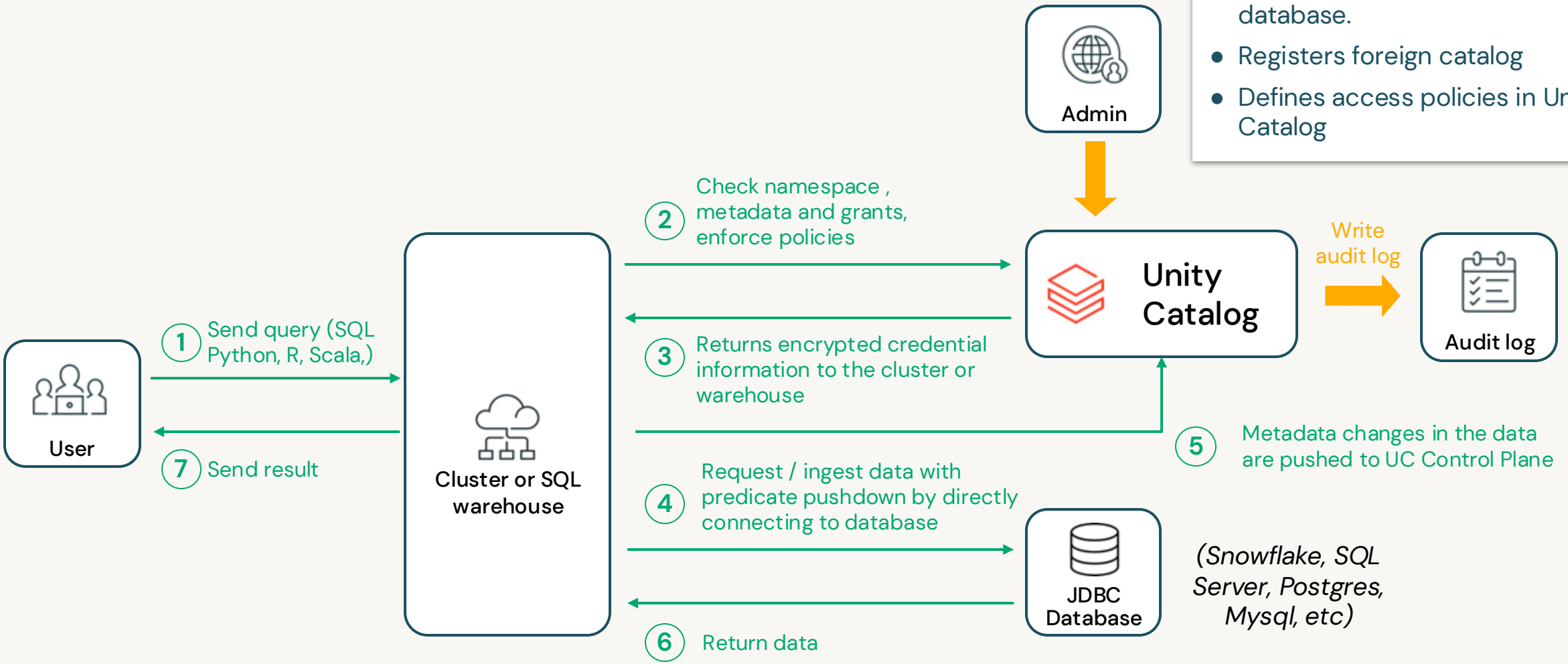
Querying file based data sources with Unity



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Querying database sources with Unity

- Defines Connection object with JDBC connection information to database.
- Registers foreign catalog
- Defines access policies in Unity Catalog



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Volumes in Unity Catalog

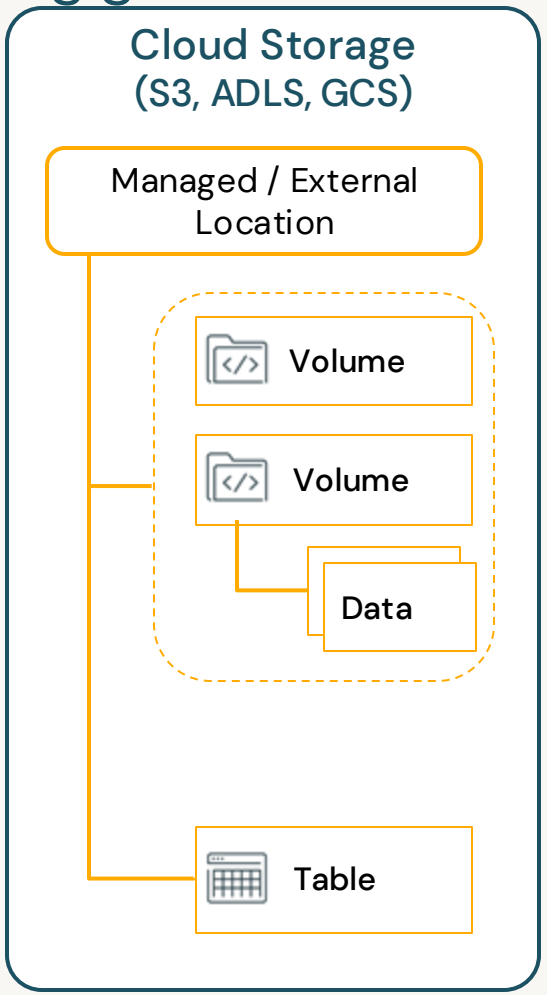
Access, store, organize and process files with Unity Catalog governance

- Volumes can be accessed by POSIX commands

```
dbutils.fs.ls("s3://my_external_location/Volumes/volume123")

ls /Volumes/volume123
```

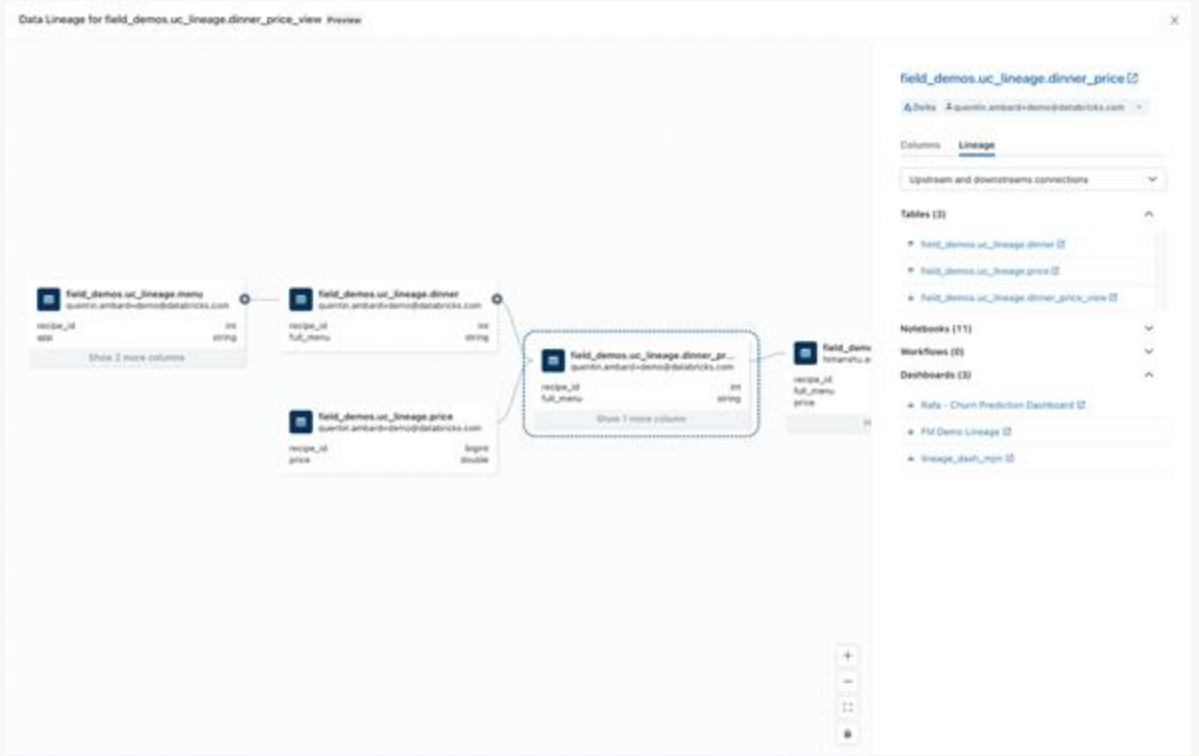
- Volumes are created under Managed or External Locations and show up in UC Lineage
- Volumes add governance over non-tabular data sets
  - Unstructured data, e.g., image, audio, video, or PDF files, used for ML
  - Semi-structured training, validation, test data sets, used in ML model training
  - Raw data files used for ad-hoc or early stage data exploration, or saved outputs
  - Library or config files used across workspaces
  - Operational data, e.g., logging or checkpointing output files
- Tables are registered in Managed / External Locations, not in Volumes



Automated lineage for all workloads

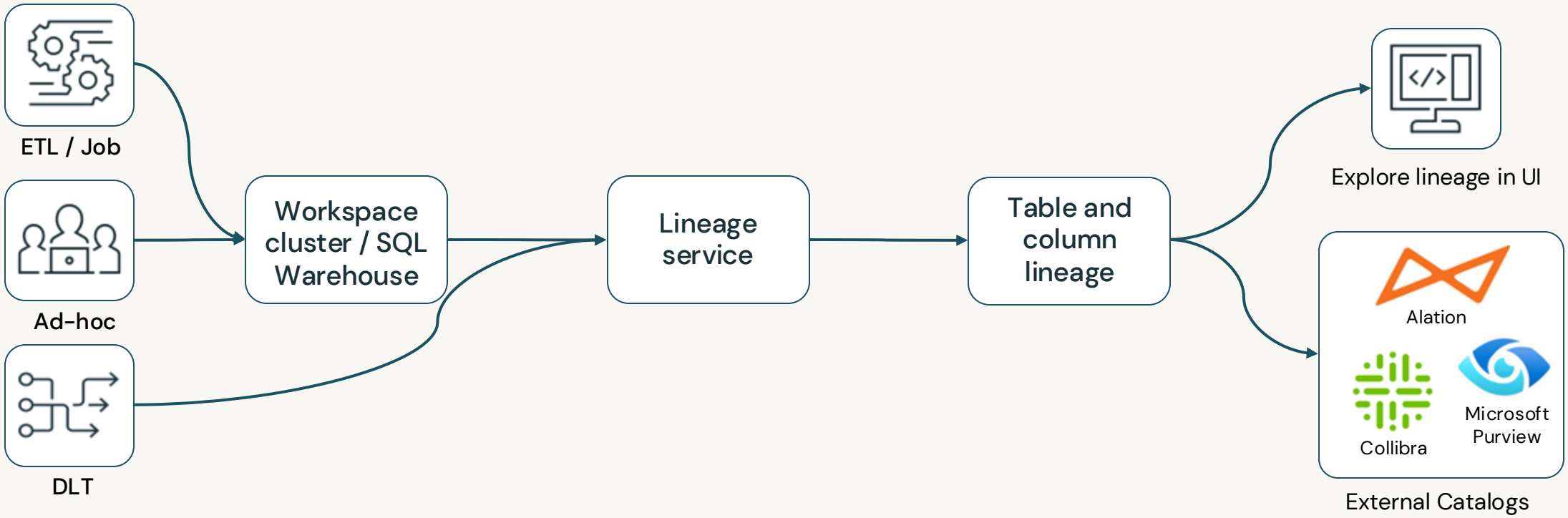
End-to-end visibility into how data flows and consumed in your organization

- Auto-capture runtime data lineage on a Databricks cluster or SQL warehouse
- Track lineage down to the table and column level
- Leverage common permission model from Unity Catalog
- Lineage across tables, dashboards, workflows, notebooks, feature tables, files, and DLT



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Lineage flow – How it works



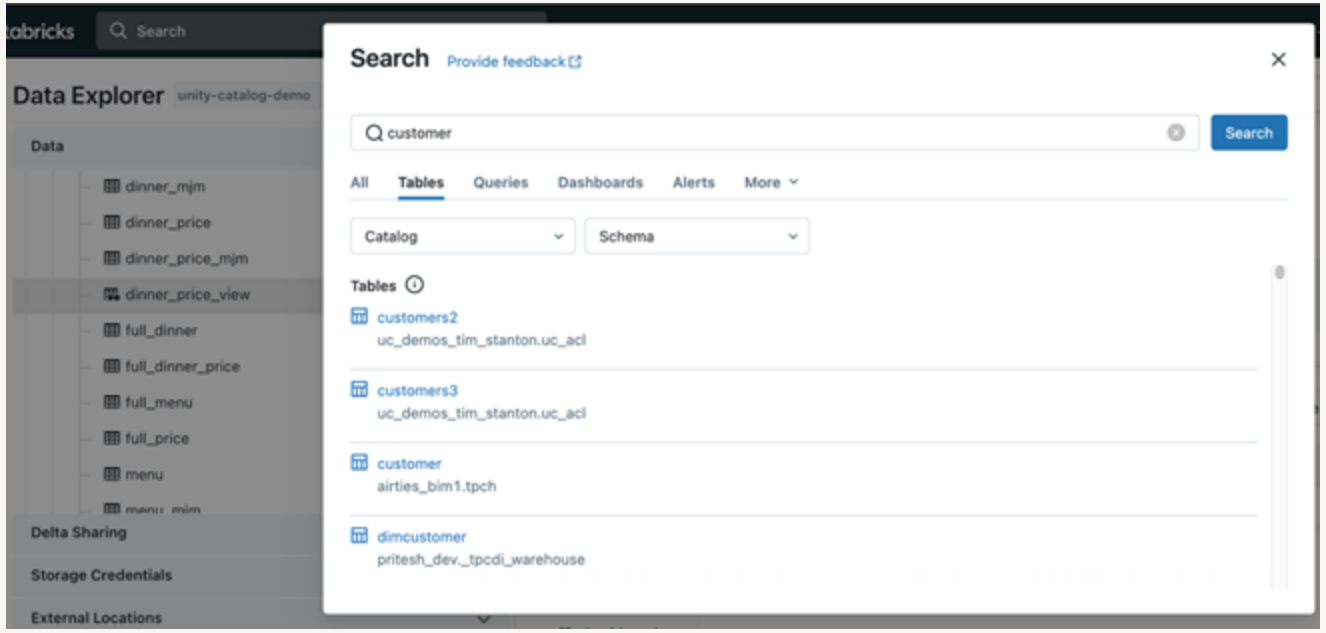
- Code (any language) is submitted to a cluster or SQL warehouse or DLT* executes data flow
- Lineage service analyzes logs emitted from the cluster, and pulls metadata from DLT
- Assembles column and table level lineage
- Presented to the end user graphically in Databricks
- Lineage can be exported via API and imported into other tool

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Built-in search and discovery

Accelerate time to value with low latency data discovery

- UI to search for data assets stored in Unity Catalog
- Unified UI across DSML + DBSQL
- Leverage common permission model from Unity Catalog
- Apply semantic tags to data and search across tags



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Discovery Tags

Semantic layer for your lakehouse

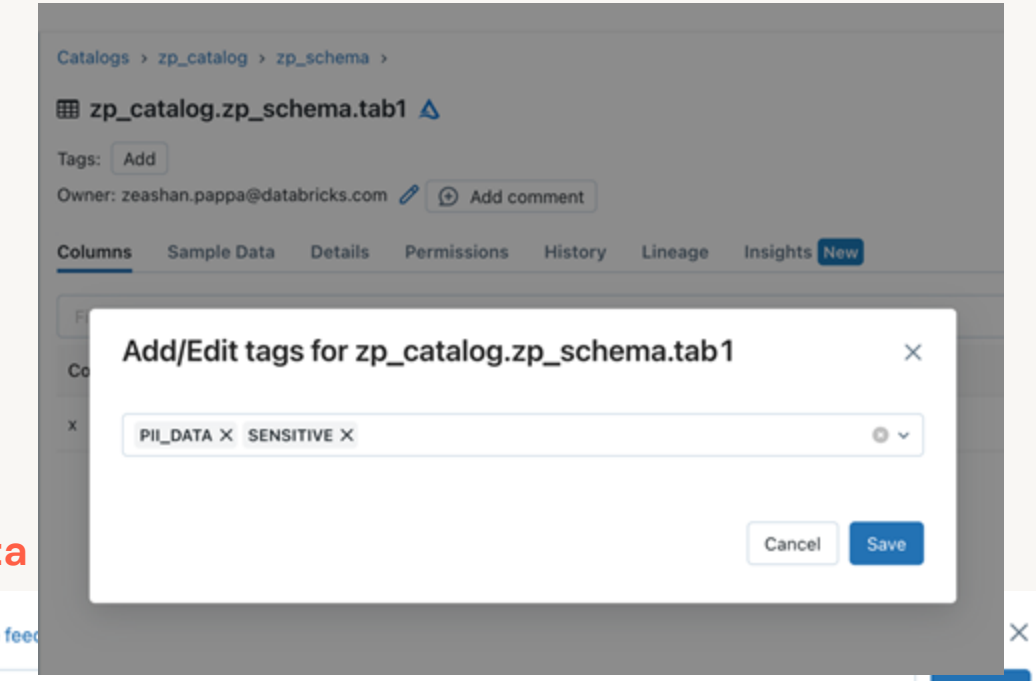
Problem

Searching for data assets in business terms or generally agreed upon taxonomies usually requires additional catalog tools.

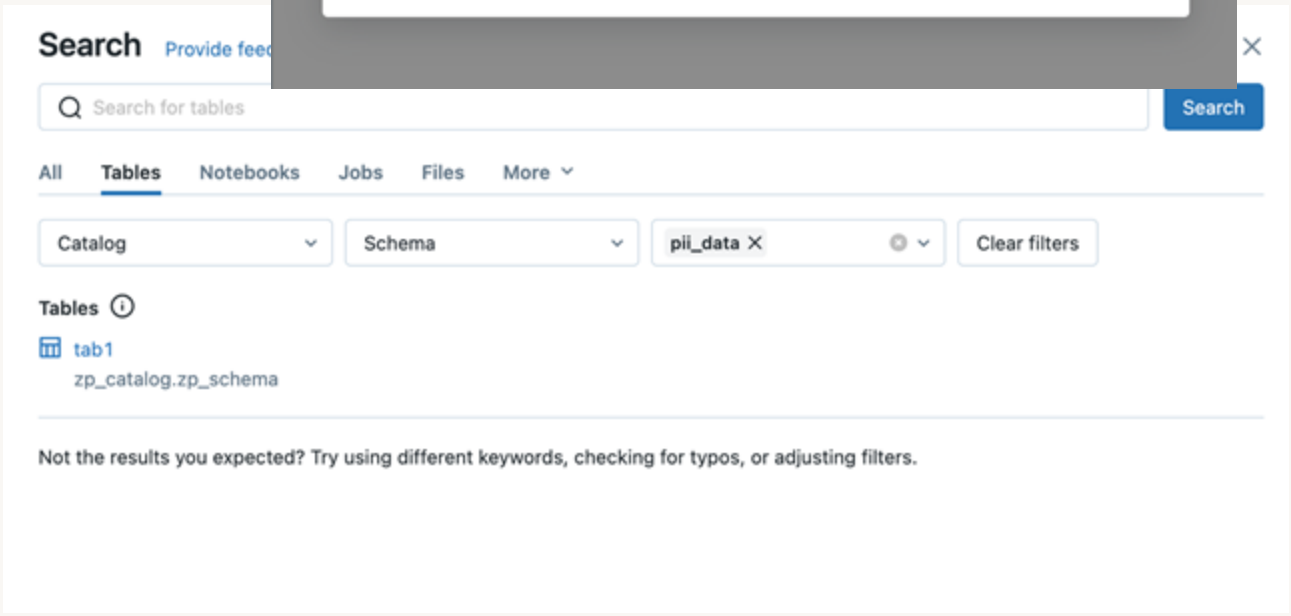
Solution

- Discovery Tags allow you to tag Column, Table, Schema, Catalog objects in UC
- **Integrated search mechanism in UC allows you to search for objects by tag.**

Tag your data

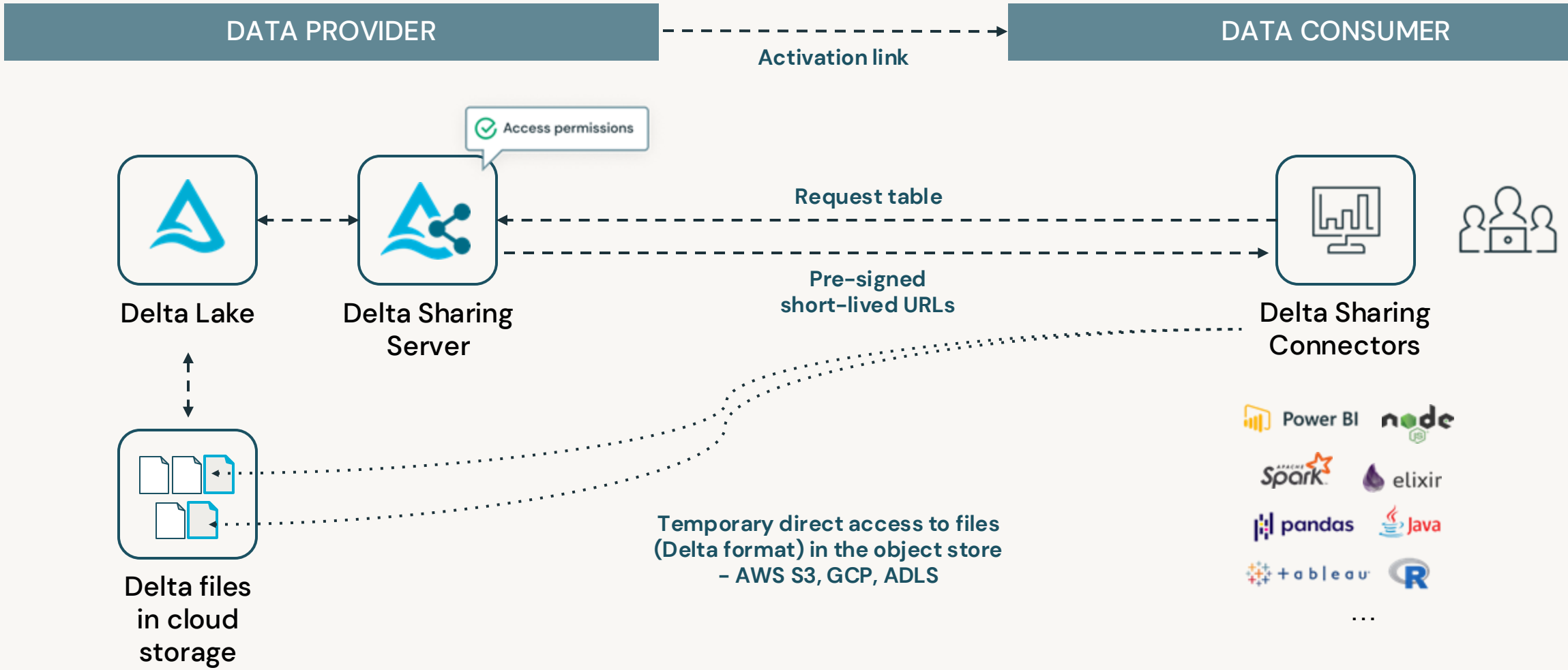


Search based on tags



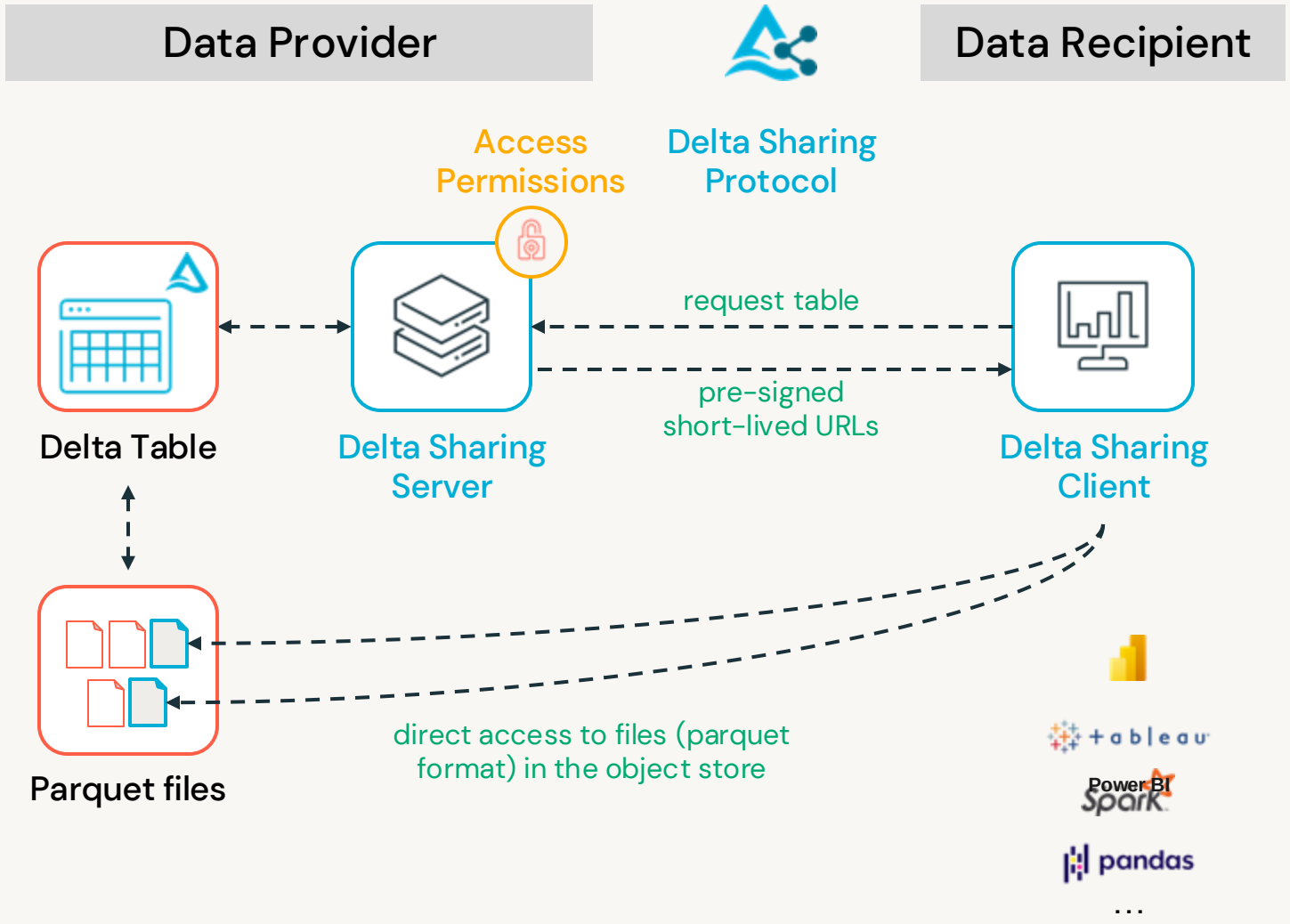
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Delta Sharing



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Delta Sharing – Under the hood



Delta Sharing Protocol:

- Client authenticates to Sharing Server
- Client requests a table (including filters)
- Server checks access permissions
- Server generates and returns pre-signed short-lived URLs
- Client uses URLs to directly read files from object storage

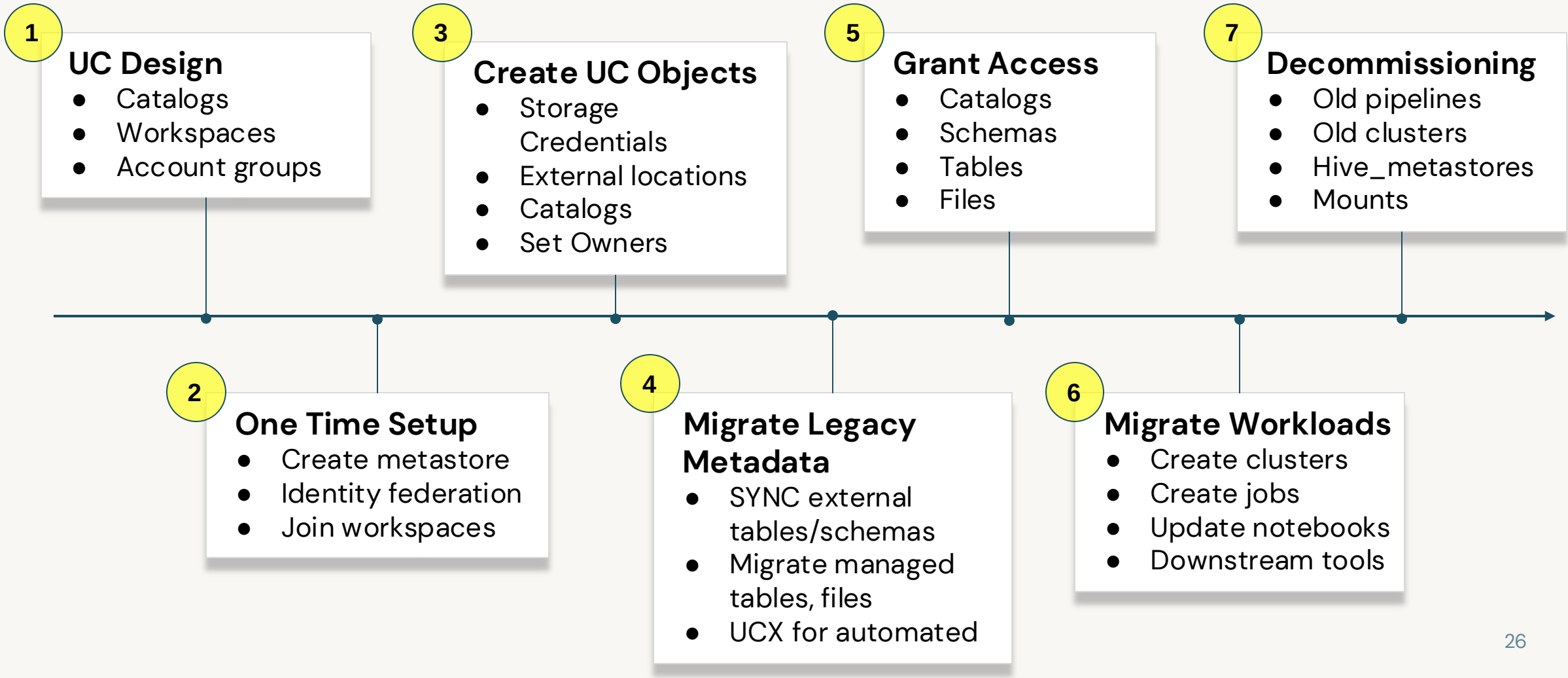
Notes:

- Sharing happens on Delta part files, supporting full tables, partitions, delta versions, ...
- Client is system independent, just needs to be able to read parquet files
- In Databricks Sharing Server and ACL checks are integrated with Unity Catalog

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High Level Roadmap for HMS to UC Migration

Steps to consider



Thank you

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